



Society of Petroleum Engineers

SPE-227331-MS

Revolutionizing Localization Towards a New Era of Sustainability

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

This project establishes a local chemical manufacturing facility for the organization to meet market dynamics, align with the localization vision, and reduce carbon emissions. It minimizes the environmental footprint by transitioning from international suppliers, enhancing In Kingdom Total Value added (IKTVA) score, and fostering technological advancements. The project aims to optimize lead times, reduce CO₂ emissions, meet financial metrics, improve compliance, and enhance business continuity through market analysis, process optimization, Lump sum Turnkey (LSTK) operations, and improved efficiency.

The project's strategy encompasses three core elements. Firstly, it involves establishing a local chemical manufacturing facility through a collaboration with our Joint venture in country. This partnership leverages existing manufacturing infrastructure, optimizing resources utilization. Secondly, it focuses on forming partnerships with established market players to facilitate technology transfer. This eliminates the need to build processes from scratch, saving valuable time and resources. Thirdly, the project entails the development of a dedicated research and development team. The goal is to create solutions for specific field requirements while ensuring the timely delivery of high-quality materials.

The localization of specialty chemicals manufacturing has yielded substantial benefits for the organization. By integrating backward into chemical manufacturing, operational efficiency enhanced, leading to a 15% reduction in product costs through supply chain optimization. Over the 2022-2027 forecast period, a remarkable 330% growth in local spending is anticipated. Conversely, international spending is projected to decline by 51%. This shift will elevate local spending, signifying the project's success in promoting localized production. In terms of business continuity, localizing specialty chemicals has mitigated supply chain disruptions, bolstering the delivery capabilities and ensuring consistent market demand fulfillment. Financially, the localization initiative has been fruitful, with exports driving substantial financial growth. Through this initiative we have successfully exported specialty chemicals to countries like to multiple geographies inside and outside the middle east, expanding its global presence. Furthermore, the introduction of localized products has positively impacted the business. Localized chemicals like Surfactants, Shale Inhibitors, Emulsifier, and Corrosion Inhibitors have contributed to a sustainable way of work. Additionally, product portfolio is expanded to include over 30 varieties, such as Viscosifiers, Rheological Modifiers, HPHT Fluid Loss Control. Aiming to further enhance its competitive edge and market position.

The localization project responds to dynamic market shifts while championing sustainability. It aligns with localization goals, reinforcing partnerships and shared sustainability targets. Environmental sustainability is paramount, with significant carbon emissions reduction through local manufacturing. In essence, localization project signifies the commitment to sustainability, efficient operations, and long-term industry success.

Introduction

The global oil and gas industry is experiencing a transformative era, marked by the increasing need for sustainability, cost efficiency, and reduced dependence on international supply chains. In this dynamic environment a Global Supplier (GS) in the oil and gas sector, has recognized the imperative to establish a local chemical manufacturing facility in a central operation in Middle East. This paper explores the strategic initiative between a global leader in oilfield services company and a local manufacturing company in localizing specialty chemical production.

The initiative is not only a strategic move to adapt to evolving market dynamics but also a commitment to enhancing business sustainability and operational efficiency. This paper will detail the motivations, processes, and impacts of this localization effort, highlighting its significance for all involved parties as well as the Middle East oil and gas industry.

The Need for Local Manufacturing Facility

A major player in Middle East oil and gas sector, faced several challenges in maintaining its market position and financial viability of projects execution. The landscape, increasingly dominated by local manufacturers, demanded a strategic shift. Three key factors drove the decision to establish a local manufacturing facility:

- **Cost-Efficiency:** Relying on international suppliers for specialty chemicals resulted in higher production costs. Local manufacturing presented an opportunity to reduce these costs, thereby enhancing competitiveness in a cost-sensitive market.
- **Supply Chain Disruptions:** The logistical complexities and prolonged lead times of importing chemicals internationally posed significant risks to operational efficiency and financial liquidity. The recent disruptions in the Red Sea shipping routes, a critical transit point for international supplies, further underscored the vulnerability of international supply chain to geopolitical and regional instabilities. Establishing a local facility aimed to mitigate these risks, ensuring a more reliable and efficient supply chain.
- **Environmental Impact:** The transportation of chemicals over long distances contributed substantially to carbon emissions. By localizing production, carbon footprint is reduced, aligning with company commitment to environmental stewardship and sustainability.

Main Customer Support for Local Manufacturing Units

The main customer, as a key stakeholder, played a crucial role in supporting local manufacturing initiatives. Customer endorsement to localization project aligned with several of its strategic objectives:

- **Sustainability:** particularly the ‘In-Kingdom spend by 2030’ initiative, prioritize responsible and environmentally conscious sourcing. Supporting local manufacturing was a direct reflection of this commitment.
- **Supply Chain Efficiency:** By sourcing locally, Customer anticipated improvements in supply chain efficiency, crucial for the seamless operation of its projects.
- **Cost Optimization:** Local sourcing was seen to reduce the costs associated with importing specialty chemicals, thereby enhancing Customer's profitability.

- **Local Economy:** Supporting local manufacturing units resonated with Customer's broader economic diversification goals, fostering job creation and stimulating the local economy.

Considering these, a strategic journey was established by the GS with a local chemical manufacturing company (LC), marking a significant shift in GS operational strategy.

Overview of the Local Manufacturing Sector for Oil and Gas Specialty chemical

Saudi Arabia's manufacturing sector, particularly in oil and gas, is transforming in line with Vision 2030 and the National Industrial Development and Logistics Program. Central to this vision is Saudi Aramco's aim to increase localization in the Oil and Gas sector from 45% to 70% by 2030. This strategic shift is driving investments in local specialty chemical manufacturing facilities, supported by government incentives and collaborations with international industry leaders. These efforts focus on innovation and sustainability while addressing challenges like skill development and market adaptation. As a result, Saudi Arabia's specialty chemicals sector is poised for growth, positioning the country as a global player in specialty chemicals.

Government Policies and Initiatives

IKTVA Program: A Catalyst for Localization

The Saudi government's IKTVA program (<https://iktva.sa/>), launched as part of Vision 2030, has been instrumental in reshaping the industrial landscape of the Kingdom of Saudi Arabia (KSA), particularly in the oil and gas sector. Designed to promote economic diversification and reduce dependence on oil revenue, IKTVA has focused on incentivizing local manufacturing to bolster the national economy.

The IKTVA program's incentives, including tax benefits, subsidies, and regulatory support, were pivotal in the Global Supplier (GS) decision to partner with the Local Manufacturing Company (LC) for establishing a local specialty chemical manufacturing facility. The program's emphasis on local value creation aligned seamlessly with GS strategic objectives, facilitating the company's transition from an international supplier to a local manufacturer.

Strategic Impact of Government Incentives

The Saudi government's incentives under the IKTVA program were not merely financial boosts; they represented strategic tools that enabled GS to realign its operational model. Tax benefits and subsidies reduced the financial risk associated with the establishment of new manufacturing facilities. Meanwhile, regulatory support streamlined the process of technology transfer and talent acquisition, critical components of the localization strategy.

Moreover, the program's focus on local content development dovetailed with GS efforts to integrate into the Saudi industrial fabric. By complying with the program's requirements, GS not only benefited from the government's incentives but also strengthened its relationship with key stakeholders, including Saudi Aramco, which increasingly preferred local manufacturers for its supply chain.

Long-term Implications for the Sector

The IKTVA program signifies a paradigm shift in Saudi Arabia's industrial policy, with long-term implications for international companies. By fostering an environment conducive to local manufacturing, the program has encouraged the development of a robust, diversified, and self-sufficient industrial base in the Kingdom.

For GS, this shift meant reevaluating its global supply chain strategy, emphasizing local production and talent development. The localization initiative, thus, is not merely a response to a government policy but a strategic realignment with the future trajectory of the Saudi economy.

The Strategic Entry in Localized Specialty Chemical Sector

The Evolution of Saudi Arabia's Manufacturing Landscape

The Kingdom of Saudi Arabia, traditionally reliant on oil revenues, has witnessed a significant transformation in its industrial landscape over the past decade. This change, catalyzed by Vision 2030, aimed to diversify the economy and reduce oil dependency. The oil and gas sector, however, remains a cornerstone of the Saudi economy, providing a fertile ground for the growth of ancillary industries, including specialty chemical manufacturing.

The local manufacturing sector, once nascent, has burgeoned, driven by government initiatives and the entry of multinational corporations. The sector's growth is not just reflected in the increased number of manufacturing units but also in the sophistication and range of products being produced. Specialty chemicals for the oil and gas industry represent a notable segment of this growth, driven by the demand from major companies like Saudi Aramco.

Global Supplier Strategic Entry into Local Manufacturing

The Global Supplier (GS) foray into the local manufacturing sector in Saudi Arabia was a strategic decision influenced by several factors. The competitive landscape in the oil and gas sector was rapidly changing, with a growing preference for local suppliers. This shift was partly due to the cost and efficiency advantages local manufacturers offered, aligning with Aramco's emphasis on local sourcing.

Market Entry Strategy and Investment

The Global Supplier (GS) market entry strategy was characterized by a substantial investment in technology and infrastructure, partnering with the Local Manufacturing Company (LC). This collaboration aimed to leverage local expertise and global experience. The magnitude of the investment reflected the global supplier's commitment to the region and its long-term vision for its role in the local economy.

Technological Advancements in Local Manufacturing

Collaboration with Global Technology Leaders

The technological leap achieved by the GS was facilitated through strategic collaborations with global technology leaders. These partnerships allowed for the transfer of technical know-how and the adaptation of international best practices to the local context.

One notable aspect of these collaborations was the focus on sustainable and environmentally friendly technologies. This aligns with both the GS corporate sustainability goals and Saudi Arabia's Vision 2030 environmental objectives. For instance, the implementation of energy-efficient systems and waste reduction techniques has set new industry standards in Saudi Arabia.

Impact on Manufacturing Efficiency and Sustainability

The technological advancements implemented by the GS had a profound impact on the efficiency and sustainability of the manufacturing process. Enhanced efficiency was evident in reduced production times, lower costs, and improved product quality. Sustainability gains were achieved through reduced energy consumption, minimal waste generation, and a smaller carbon footprint.

Challenges and Opportunities in Localization

Navigating Challenges in Technology Importation

The journey to establish a local manufacturing base in Saudi Arabia was not without its challenges. One of the primary hurdles was the importation of advanced technologies. Navigating the regulatory landscape,

ensuring compliance with both Saudi and international standards, and managing logistical complexities posed significant challenges.

Leveraging Opportunities for Growth and Market Expansion

The localization initiative presented numerous opportunities. The most significant was the potential for market expansion within the Kingdom and the broader Middle East region. By establishing a local manufacturing presence, The Global Supplier positioned itself strategically to capitalize on regional growth opportunities, particularly in light of Saudi Aramco's preference for local suppliers.

Furthermore, the initiative opened doors for the global supplier to diversify its product offerings. The proximity to its key market allowed for better alignment with customer needs and quicker adaptation to market changes. This agility in responding to market demands became a critical competitive advantage.

Long-Term Strategic Benefits

The long-term strategic benefits of localizing specialty chemical manufacturing for Global supplier (GS) were clear. It enhanced the company's operational resilience, reduced dependency on international supply chains, and aligned it with national economic objectives. Moreover, it solidified the global supplier commitment to Saudi Arabia, enhancing its reputation and strengthening relationships with key stakeholders.

Environmental Impact and Sustainability Initiatives

Reduction in Carbon Emissions through Localization

One of the most significant environmental impacts of the localization initiative is the substantial reduction in carbon emissions. By establishing a local manufacturing facility, the need for long-distance transportation of specialty chemicals from international suppliers has been drastically reduced. This reduction in transportation not only decreases carbon emissions but also aligns with Saudi Arabia's Vision 2030 goals for environmental stewardship.

To assess the overall reduction of CO₂ emission we used a previously described methodology (Gilbert, 2010) where the transportation impact is properly quantified based on the transportation method used. In such methodology, each metric ton transported by sea freight and land transported (truck) have its CO₂ emissions quantified by kilometer transported.

We used one standard shipment of a flat rheology oil-based system (Friedheim, 2021) to quantify the impact of this nationalization effort with the following amounts. The CO₂ footprint reduction is in excess of 99% on emissions related to transportation.

Table 1—Table Summary of CO₂ footprint reduction due to localization

Product	Shipment (tons)	All Imported (KgCO ₂ /tkm)	1st Step Localization (KgCO ₂ /tkm)	2nd Step Localization (KgCO ₂ /tkm)
Emulsifier	320	104,188	675	675
Wetting Agent	80	26,047	169	169
Rheological Modifier	20	7,441	42	42
Organophilic Clay	140	26,298	26,298	295
Organophilic Clay	360	22,265	22,265	759
Total	920	186,241	49,449	1,939

Implementation of Sustainable Manufacturing Practices

The global supplier has integrated sustainable practices into its manufacturing processes in Saudi Arabia. These practices include the use of energy-efficient machinery, waste reduction techniques, and the adoption of renewable energy sources where feasible. These efforts demonstrate the commitment to minimizing the environmental footprint of its operations and contributing to a more sustainable industry.

Life Cycle Analysis of Specialty Chemicals

In addition to implementing sustainable manufacturing practices, the global supplier has conducted a life cycle analysis of its specialty chemicals. This analysis assesses the environmental impact of these products from production to disposal, ensuring that every stage is as environmentally friendly as possible. The results of these analyses have informed the development of new, more sustainable chemical formulations.

Aligning with Global Sustainability Goals

The Global supplier efforts in environmental sustainability extend beyond local compliance and align with broader global sustainability goals. The company actively participates in global dialogues on sustainability and incorporates international best practices into its local operations. This global-local approach ensures that the environmental initiatives in Saudi Arabia are part of a larger, cohesive strategy towards sustainability.

Market Trends and Future Outlook for Local Manufacturing

The specialty chemical segment within the oil and gas industry in Saudi Arabia has been evolving rapidly, influenced by global and regional market trends. One key trend is the increasing demand for environmentally friendly and sustainable chemical solutions, driven by global environmental concerns and regulatory changes. Additionally, there is a growing emphasis on cost-efficiency and supply chain resilience, further shaping the market dynamics.

The adoption of advanced technologies in manufacturing processes is another significant trend. The integration of digital technologies like AI, IoT, and big data analytics is transforming manufacturing operations, leading to increased efficiency, reduced downtime, and better-quality control. The early adoption of these technologies in its Saudi operations has allowed the GS to stay ahead in the market.

Geopolitical factors, such as regional stability and international trade policies, continue to impact the oil and gas industry. The global supplier decision to localize manufacturing in Saudi Arabia offers a strategic advantage in this context, reducing dependency on international supply chains and mitigating risks associated with geopolitical uncertainties.

Future Outlook and Growth Opportunities

Looking ahead, the future of specialty chemical manufacturing in Saudi Arabia appears promising. With the continued support of government initiatives like IKTVA and Vision 2030, along with increasing demand from major players like Saudi Aramco, the sector is poised for growth. Global supplier's established local presence positions it well to capitalize on these opportunities.

The potential for export to neighboring countries also presents a significant growth revenue for the GS. Leveraging its local manufacturing base, GS can expand its market reach, contributing to the Kingdom's broader economic diversification goals.

Impact on Global Supplier

Enhanced Local Expenditure and Economic Contribution

The localization of specialty chemical manufacturing in Saudi Arabia has significantly increased the GS local expenditure, showcasing a substantial commitment to supporting the local economy. In 2023 alone, the local spend reached approximately \$30 million. This investment not only demonstrates the GS alignment

with Saudi Arabia's economic diversification goals but also reinforces its role as a key player in the country's industrial development.

Equity Pickup and Financial Performance

The global supplier financial performance has been positively impacted by the localization initiative. The company has seen a substantial increase in equity pickup (EPU) as a result of introducing new chemicals manufactured through the joint venture with the LC. In 2024, The GS EPU reached approximately \$4.5 million, indicating the financial viability and success of the localization strategy.

Pricing Premium and Competitive Advantage

By manufacturing locally, GS has been able to command a pricing premium of 10-15% over international competitors. This pricing advantage has bolstered the GS market position, contributing to increased profitability and market share within the Kingdom.

Job Creation and Social Contributions

The establishment of the local manufacturing facility has led to the creation of over 40 new jobs in Saudi Arabia under the local supplier partnership, fostering economic growth and offering valuable employment opportunities. This contribution aligns with the GS corporate social responsibility goals and supports the broader economic objectives of Saudi Arabia.

Inventory Reduction and Operational Efficiency

The localization initiative has also resulted in an anticipated \$7.5 million inventory reduction for the GS in 2024. This reduction in inventory levels signifies improved operational efficiency and cost-effectiveness, crucial in a competitive industry landscape.

Conclusion: Synthesizing the Impact of Localization in the Saudi Oil and Gas Sector

The localization of specialty chemical manufacturing through a partnership between Global and Local companies marks a significant milestone in the Saudi oil and gas industry. This initiative, aligned with Saudi Arabia's Vision 2030 and the IKTVA program, has demonstrated the strategic benefits of local manufacturing in a sector traditionally dominated by international suppliers.

The localization initiative has shown a strong alignment with Saudi Arabia's national objectives of economic diversification and sustainability. The increased local expenditure, job creation, and support of local supply chains underscore global suppliers commitment to contributing to the Kingdom's economic development.

The strategic move to localize manufacturing has enhanced Global Supplier (GS) competitiveness in the Saudi market. The ability to offer locally manufactured specialty chemicals has not only provided a pricing advantage but also positioned GS as a key player in the regional oil and gas sector. The initiative has improved GS market access, enabling it to better serve its primary client and expand its customer base.

The environmental benefits of this localization initiative are noteworthy. The reduction in carbon emissions due to decreased transportation needs and the adoption of sustainable manufacturing practices reflect GS commitment to environmental stewardship. These efforts align with global sustainability goals and contribute to a more responsible industry.

The success of this localization initiative opens avenues for further research and collaboration in the field of local manufacturing. It sets a precedent for other multinational corporations considering similar ventures in Saudi Arabia and the broader Middle East region.

In conclusion, this initiative to localize specialty chemical manufacturing in Saudi Arabia offers insights into the strategic benefits of local manufacturing. It highlights the importance of aligning with national

economic goals, adapting to market demands, and committing to sustainability. As Saudi Arabia continues to diversify its economy, the lessons from this experience will be valuable for companies looking to contribute meaningfully to this transformation.

Acknowledgments

The authors thank the global supplier leader in oilfield services company for approval of publishing this paper.

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